Ethics of AI

DIABETIC RETINOPATHY SCREENING IN LOW-RESOURCE SETTINGS

CASE BRIEF

Diabetic retinopathy (DR) is a leading cause of preventable blindness, and diagnosing it requires trained specialists to examine retinal images. In Thailand, there are roughly 1,500 ophthalmologists serving 4.5 million diabetic patients — a ratio of about 1 specialist per 3,000 patients, nearly double the shortage seen in the U.S. Thailand's Ministry of Public Health set a national goal in 2013 to screen 60 percent of diabetic patients annually for DR, but because of the shortage of trained graders, less than half of patients were actually screened most years.

In 2016, Google researchers developed a deep-learning algorithm that could assess retinal images for DR with specialist-level accuracy in lab conditions, without needing to wait weeks for an ophthalmologist's review. Between 2018 and 2019, Google Health partnered with clinics in Thailand to deploy the tool in real-world conditions across 11 sites. This deployment surfaced problems the lab testing hadn't: the tool needed high-quality retinal images to function accurately, but clinic staff couldn't always produce them — lighting conditions, camera quality, and internet connectivity varied significantly between clinics, and screening workflows differed from site to site. When an image was rejected as too low-quality for the algorithm to assess, some patients who would previously have received at least a provisional read from a nurse got no assessment at all, and had to return for another visit.

Google and its research partners continued refining the tool's integration into Thailand's national screening infrastructure over the following years. By 2022, a follow-up study evaluating the system's real-time performance across nine primary care sites within Thailand's actual national screening program found it detecting vision-threatening DR with 94.7 percent accuracy, 91.4 percent sensitivity, and 95.4 percent specificity — matching or exceeding the performance of retina specialists themselves. This marked a substantial shift from a standalone pilot to a tool operating reliably inside the country's live public health system.

By 2024, the model had already been used to screen more than 600,000 patients across clinics worldwide. That year, Google announced it was licensing the model to healthcare partners — Forus Health, AuroLab, and Perceptra — for scaled deployment across India and Thailand, with a stated goal of delivering six million AI-enabled screenings to under-resourced communities at no cost to patients over the next decade.

PART 1 — INDIVIDUAL SORTING

Sort each item into: (a) non-maleficence norm, (b) beneficence norm, (c) unclear/both. As before, the ones you mark unclear are the ones to come ready to defend.

1. The decision to deploy an AI screening tool at all in a region with a severe, well-documented shortage of eye specialists.
2. A requirement that retinal images meet a minimum quality threshold before the algorithm will assess them.

3. The finding that patients whose images were rejected for low quality had to make a second trip to the clinic, adding cost and delay to their care.
4. The commitment to deliver six million free AI-enabled screenings to under-resourced communities over ten years.
5. A design rule that the AI flags a case for a human specialist rather than issuing a final diagnosis on its own.
6. Google and its Thai partners spent several years running further studies inside the national screening program before scaling the tool nationally in 2024, rather than scaling immediately after the 2019 pilot.
7. The finding that some borderline cases led to unnecessary referrals to specialists, adding strain to the already-scarce workforce the tool was designed to support.
8. Embedding the AI into Thailand's national screening program, which had never reached its own screening target because of a shortage of trained graders.

PART 2 — SMALL GROUP DISCUSSION

1. Item 3 (the second-trip burden) only exists because the tool was deployed to help people in the first place. Is a harm that's created as a side effect of a beneficent intervention still a pure non-maleficence failure, or does it need its own category — a cost of doing good that isn't the same as doing harm? Does the answer change your view of whether the deployment was net justified?
2. Item 6 (waiting years to scale rather than moving immediately) prevents a harm, but every year the tool isn't scaled, patients who could have been screened aren't. Is choosing caution over speed here a non-maleficence win, a beneficence failure, or both at once? Is there a point where "we should have waited" becomes its own kind of harm?
3. Unlike Grok, nothing about this technology creates the harm it's trying to prevent — diabetic retinopathy exists independent of the AI. Does that mean non-maleficence has a much smaller role to play in evaluating this case than beneficence does? Or is non-maleficence just showing up in a different place (deployment quality, oversight, equity of accuracy across populations) rather than being absent?
4. Google frames the six-million-screenings commitment as free, at scale, and reaching people who currently have no access at all. Is "expanding access to something that didn't exist before" always beneficence, or could you argue it's actually a non-maleficence claim in disguise — preventing the harm of untreated, undiagnosed blindness? Does the distinction actually matter here, or does it collapse?